

Kineo SRS Document

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1. Introduction

1.1 Purpose

This Software Requirements Specifications (SRS) document provides a comprehensive description of the AI-powered travelling application intended for creating planned itineraries and optimized trip recommendations.

This SRS will serve as a foundation of the project consisting of essential system designs, diagrams, development phases, etc. ensuring a clear understanding of how the system operates and what it does.

1.2 Intended Audience

This document is for these stakeholders:

Stakeholders	Role
Customers	Individuals who will purchase and use the application.
End-users	Individuals who will use the application and provide feedback.

1.3 Product Scope

The AI-powered travelling application **Kineo** is designed to support spontaneous travellers by automating the destination discovery process. Using advanced Google Gemini Artificial Intelligence, the platform is able to provide smart budget estimations, trip planning, enhancing user's satisfactions.

In-Scope

The application will include these key concepts:

1. Security Support via Firebase: The app provides secure authentication and authorization in signing-in or signing-up.
2. Google Gemini: Developing relevant resources, data screening, data extraction, and searching strategy. How this works is Gemini uses a multi-layered search strategy wherein breaks the query into subparts (Query Decomposition) and thoroughly understands the user's input through semantic search. Secondly, it develops relevant sources by identifying relevant and trusted resource types such as Authoritative travel sites (booking.com, TripAdvisor, etc.), blogs, news, and more. Thirdly, Data Screening works efficiently by following the filter standard which evaluates each source based on relevance scoring, credibility checking, and recency filtering to ensure that only accurate, up-to-date, and trustworthy information makes it into the final output. And lastly, data extraction works by using techniques such as Named Entity Recognition, Relation Extraction, and more to pull structured facts like hotel names, prices, locations, and travel dates from unstructured web content, transforming raw text into clean, machine-readable data that can be directly used to build a complete travel plan.

3. Integration Framework: APIs used for providing location data of geographic locations.
4. Firestore Database: A database is provided storing user's information safely and with confidentiality.

Out-of-Scope

The application will not include these key concepts:

1. Payment Process.
2. Actual Booking of flights, hotels, etc.
3. Maintenance and long term updation.

1.4 Document Conventions

1. This document follows the APA 7 format of referencing.
2. This document is written in the Georgia font.

1.5 References

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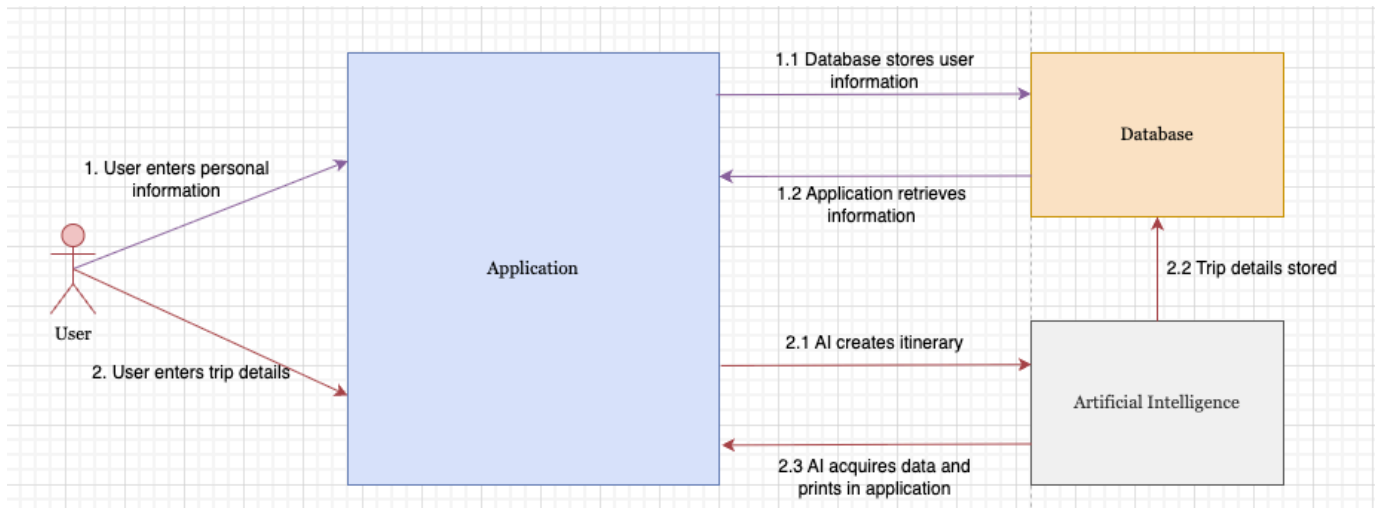
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2. Overall Description

2.1 Product Perspective

This product will operate as a standalone application to address the challenges of the travelling and booking industry. The user will enter relevant information such as destinations, dates, etc., AI converts it into an itinerary, and database stores.



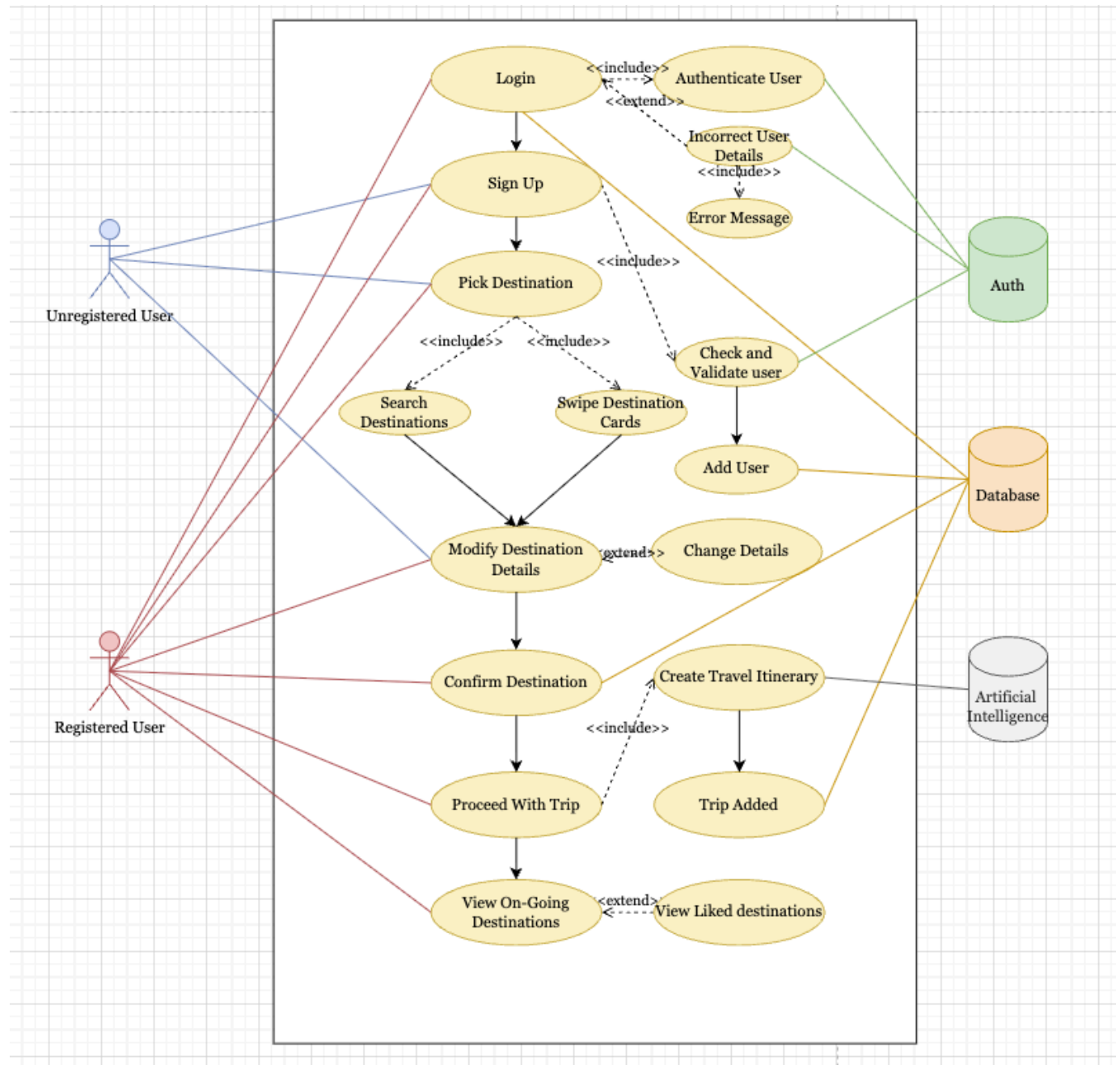
2.2 Product Function

The Kineo Application will provide such functions:

2.2.1 User Functions:

- a. User Registration
 - ❖ Traveller account registration and sign-in into the application.
- b. Login/User Information
 - ❖ Traveller's sign in information saved for future sessions.
- c. Destination Variety
 - ❖ Travellers are able to search a wide range of destinations on the app.
- d. Change Destination Information
 - ❖ Travellers can modify travel destination details.
- e. Liked Destinations
 - ❖ Travellers can 'Like' a destination and it can be accessed in a separate page.

2.3 User Classes and Characteristics



2.3.1 Registered Users

- ❖ Users that have given information such as email ID and password will be considered as registered users and are able to access destination searching, destination details, and liked destinations.

2.3.2 Unregistered Users

- ❖ Users that have not given information such as email ID and password will be considered as unregistered users and are able to access everything the registered users can with the exception of proceeding to the trip and liked destinations

2.3.3 Database

- ❖ The admin has the access to the database wherein it has the users that have been added with their respective travel information.

2.3.4 Authentication

- ❖ The application uses Firebase for authenticating/validating user email and password information as well as it is responsible for handling the user database.

2.3.5 Gemini Artificial Intelligence

- ❖ Gemini converts user travel details of a user into an AI prompt and adds essential details such as hotel recommendations, specific travel destinations, possible costs, etc. This is done through a technique called Prompt Engineering combined with structured output generation. Basically, it parses the input made by the user, extracts the important entities (destination, date, budget, etc.) and retrieves data from the internet to create a complete structured output in the form of a JSON format. The extracted entities are then passed through a data retrieval layer where Gemini pulls real-time information from trusted sources such as hotel booking platforms, tourism websites, and flight aggregators to ensure the output reflects accurate, up-to-date pricing and availability. This structured JSON output can then be seamlessly consumed by travel applications, booking engines, or other AI agents to render fully personalized itineraries, cost breakdowns, and destination guides without any additional processing.

2.4 Operating Environment

The platform will operate in the following environment:

2.4.1 Technical Environment

- ❖ A mobile application that can be run on IOS devices. But, this application is yet optimized for android use.

2.5 Design and Implementation Constraints

The following constraints will impact the design and implementation of the platform:

2.5.1 Technical Requirements

- ❖ It must have a paid subscription of AI API to process multiple (5 or more) prompts. Additionally, it must have a paid subscription of Database to store user information in the long run.

2.5.2 Regulatory Constraints

- ❖ Must adhere to copyright laws with the usage of pictures and other data.

2.5.2 Design Constraints

- ❖ Firstly, the color, font, font-size, etc. must be uniform across all pages of the application. Secondly, the User Interface must be comprehensible and understandable for all viewers.

2.6 User Documentation

The product will include a system demonstration material such as screenshots with instructions.

2.7 Assumptions and Dependencies

Assumptions

- ❖ Provides complete information such as a calendar for time ranges, budget, destinations, etc. AI will focus on destination recommendations rather than more complicated and advanced generation.

Dependencies

- ❖ Firstly, intermediate level AI only applied across the whole platform. Secondly, no paid or other APIs (Google Flights, Airline, etc.). And lastly, no real time bookings have been added as of the moment.

3. External Interface Requirements

3.1 User Interfaces

3.1.1 Login and Sign-up Page

- ❖ To use all of the application's functions, users must register with a working email address and a special password. Firebase Authentication verifies each user's information when they log in and delivers a go signal when everything is okay. User data is kept in a safe database.

3.1.2 Explore Page

- ❖ Includes a search bar where users can find destinations suitable to their own liking. In the “Popular Destinations” card, a swiping mechanism is added for pre-added locations for users. If swiped right, it will appear on the Trips page and if not, it will disappear.

3.1.3 Trips Page

- ❖ Will showcase the Liked destinations from either the swiping or the searching a destination. Users can select Liked destinations and possibly modify details. Users can also deselect trips from Liked destinations.

3.1.4 Travel Details Page

- ❖ This page highlights the key details of a trip, including a calendar feature that allows users to select their start and end dates. It also features a “Venture Trip” card where users can indicate the number of travelers (1, 2, 3–4, or 5+). Additionally, the page includes a Budget section with four distinct options, enabling users to choose from a range of budget preferences.

3.1.5 Profile Page

- ❖ This section displays the user’s personal information, including their name, email address, and password details.

3.2 Hardware Interface

To use this application, users must have an IOS device, Expo Go downloaded, and must be connected with the code itself.

3.3 Software Interface

The Kineo application integrates multiple external interfaces ensuring complete functionality, security, and data processing.

3.3.1 Firebase’s Database Service

The application uses Firebase Firestore database to handle storage and retrieval of user data.

- ❖ Stores user email and password securely and it maintains user’s trip and itinerary data.

3.3.2 Firebase’s Authentication Service

Firebase manages user registration and login processes.

- ❖ It handles secure email and password verification and ensures encrypted information storage.

3.3.3 Google AI API

The system connects an external AI in handling user personalized travel recommendations.

- ❖ It converts user input (searched destination, dates, people, etc.) into structured AI prompt and generates appropriate budget estimations and provides location based suggestions.

3.3.4 Location Service API

Uses Google's readily available API for retrieving geographic data and destinations.

- ❖ Fetches place names and details.

3.3.5 Development Framework

The application itself is run in Expo Go, enabling testing and possible deployment for IOS devices.

- ❖ It supports React Native development and it allows real time updation during development for easier error handling.

4. System Features

4.1 Use Case Name - Login

4.1.1 Objective - Verifies registered users information and sends a signal if all essential information is given correctly.

4.1.2 Priority - This objective is of high priority.

4.1.3 Actors - Registered Users and Database

Use Case Name	Login
Trigger	The user wants to login into the application.
Pre-Condition	The application must be opened and must be in the login/sign-up page.
Scenario	1. The user signs up by entering an email account and a valid password. 2. The user clicks the "Login" button. 3. Firebase Auth authenticates user information. Alternative Path 3.1. If email account and password has been incorrectly entered - The system displays an error message. - The user will try again.
Post-Condition	The user successfully enters the application.

4.2 Use Case Name - Sign Up

4.2.1 Objective - User signs in into an existing account.

4.2.2 Priority - This objective is of high priority.

4.2.3 Actors - Registered Users, Unregistered Users, and Database.

Use Case Name	Sign Up
Trigger	The user wants to sign up into the application.
Pre-Condition	The application must be opened and must be in the login/sign page.
Scenario	1. The user enters an email account and a valid password. 2. The user clicks the “Sign-Up” button. 3. Firebase Auth authenticates user information and Database adds new user information.
Post-Condition	The user logs in and successfully enters the application.

4.3 Use Case Name - Search Places

4.3.1 Objective - Users search for a specific place on the search bar.

4.3.2 Priority - This objective is of medium priority.

4.3.3 Actors - Registered Users, Unregistered Users, and Database.

Use Case Name	Search Places
Trigger	The user clicks the search button and searches a place of destination.
Pre-Condition	Users must be in the explore page.
Scenario	1. User clicks the search button 2. User searches and options of places will automatically show.

Post-Condition	Users will be redirected to the Trip Details page.
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4.4 Use Case Name - Modify Destination

4.4.1 Objective - In the Trip Details page, users can modify the trip details(budget, people, date).

4.4.2 Priority - This objective is of high priority.

4.4.3 Actors - Registered Users, Unregistered Users, and Database.

Use Case Name	Modify Destination
Trigger	After searching a destination, users can modify the destination details.
Pre-Condition	Users must search for a destination first.
Scenario	1. User clicks the search button 2. User searches and options of places will automatically show. 3. Users modify relevant information. Alternative Path 3.1 Users can change trip details in the Trip Review Page.
Post-Condition	Users will be redirected to the Trip Review page.

4.5 Use Case Name - Confirm Destination

4.5.1 Objective - User confirms the destination of choice.

4.5.2 Priority - This objective is of high priority.

4.5.3 Actors - Registered User and Database

Use Case Name	Add Destination
Trigger	The user confirms the destination
Pre-Condition	The user must the trip details completed
Scenario	1. User clicks the search button 2. User searches and options of places will automatically show.

	3. User modifies relevant information. 4. User confirms destination.
Post-Condition	User will be redirected to the Trips Page.

4.6 Use Case Name - Proceed To Trip

4.6.1 Objective - User proceeds with trip, awaiting for AI itinerary creation.

4.6.2 Priority - This objective is of medium priority.

4.6.3 Actors - Registered User and Artificial Intelligence

Use Case Name	Proceed To Trip
Trigger	The user confirms destination details and clicks the “Proceed” button.
Pre-Condition	User completes all destination details.
Scenario	1. User confirms destination. 2. User clicks the “Proceed” button. 3. Artificial Intelligence creates a detailed prompt based on the user's details.
Post-Condition	AI forwards prompt and full details will automatically show in the Trips Page.

4.7 Use Case Name - View On-Going Destinations

4.7.1 Objective - To check on-going details in the Trip Page.

4.7.2 Priority - This objective is of medium priority.

4.7.3 Actors - Registered Users

Use Case Name	View On-Going Destinations
Trigger	User enters Trips Page to view On-Going Destinations
Pre-Condition	The user must have a trip confirmed.

Scenario	1. User confirms a destination. 2. User checks the destination in the Trips Page.
Post-Condition	The user successfully checks details.

4.8 Use Case Name - Trip Added

4.8.1 Objective - Trip details are added to the database with user's information.

4.8.2 Priority - This objective is of high priority.

4.8.3 Actors - Database

Use Case Name	Trip Added
Trigger	User details and AI prompt sends data to database
Pre-Condition	Details of user and AI must be completed
Scenario	1. User adds trip details. 2. AI generates detailed prompt. 3. AI creates a full itinerary.
Post-Condition	Itinerary is created and added to the database.

4.9 Use Case Name - Refresh Destination Cards

4.9.1 Objective - Destination cards to refresh and display new places.

4.9.2 Priority - This objective is of medium priority.

4.9.3 Actors - Registered Users & Unregistered Users

Use Case Name	Refresh Destination Cards
Trigger	User clicks the refresh button
Pre-Condition	Users must be in the explore page.

Scenario	1. Pre-added cards are finished. 2. User clicks the refresh button
Post-Condition	10 new destination cards are added.

5. Non-Functional Requirements

These requirements define the quality attributes and constraints that the system must satisfy and meet.

5.1 Security Requirements

- ❖ The system must ask for authentication of the user in order to access the application's features.
- ❖ Authentication is managed securely in Firebase Authentication.

5.2 Safety Requirement

- ❖ The application must not have a malware, virus, or any harmful and malicious intent added into the systems.

5.3 Data Confidentiality

- ❖ Any information given by the user will be stored in the database securely and only authorized administrators can access such.

5.4 User Interface

- ❖ The application must provide a user-friendly, readable, consistent design to ensure positive user experience.

6. Glossary

Term	Description
Application Programming Interface (API)	A set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality.

Artificial Intelligence (AI)	Artificial Intelligence refers to computer systems or machines designed to simulate human intelligence to perform tasks like learning, reasoning, problem-solving, perception, and language understanding. It is a field of computer science that enables technology to analyze data, identify patterns, and act independently.
User Experience (UX)	Covers a user's overall experience with the product, website, or application.
User Interface (UI)	The interactivity, look, and feel of a product screen, web page, or application.